

Tooling.

- Tools: Python 3, pandas, lingua (language detection), wordcloud, matplotlib.
- Numbers: generated from saved data files, none typed by hand.
- Code and write-up: drafted with Claude Code.

1 Data collection

- Topic: K-pop boy group vs. girl group.
- Boy group: BTS *Dynamite*.
- Girl group: BLACKPINK *Kill This Love*.
- Data: YouTube Data API

Step	R function	Python function
1. Stats and details	<code>get_stats</code> , <code>get_video_details</code>	<code>p1_collect.fetch_video</code>
2. Comments	<code>get_comment_threads</code> , <code>write.csv</code>	<code>p1_collect.fetch_comments</code>
3. Language	none (not in the R script)	<code>common.detect_language</code> with <code>lingua</code>
4. Terms	<code>Corpus</code> , <code>DocumentTermMatrix</code> , <code>colSums</code>	<code>common.clean_terms</code> , <code>common.count_terms</code>
5. Word cloud	<code>wordcloud</code>	<code>p2_wordcloud.make_cloud</code>
6. Comparison	none	<code>p3_compare</code>

Table 1: Each R step of the assignment and its Python step.**1.1 Videos and statistics**

- Table 2: both videos, as of 2026-10-04.
- Fetched: top-level comments only.
- English kept: 1,000 per video, for the English clouds.

	BTS <i>Dynamite</i>	BLACKPINK <i>Kill This Love</i>
Title	BTS (방탄소년단) 'Dynamite' Official MV	BLACKPINK - 'Kill This Love' M/V
Channel	HYBE LABELS	BLACKPINK
Video ID	gdZLi9oWNZg	2S24-y0Ij3Y
Published	2020-08-21	2019-04-04
Views	2,146,874,108	2,243,728,665
Likes	39,661,294	26,742,200
Comments (YouTube total)	15,898,150	2,500,163
Comments fetched	1,965	2,127
English comments kept	1,000	1,000
As of	2026-10-04	2026-10-04

Table 2: Video details and statistics.

1.2 Sample

- Order: YouTube’s “Top comments” (relevance) first, then newest first.
- Stop: 1,000 English comments per video, or the page limit.
- Language label: Figure 1.
- Counts: Table 3.

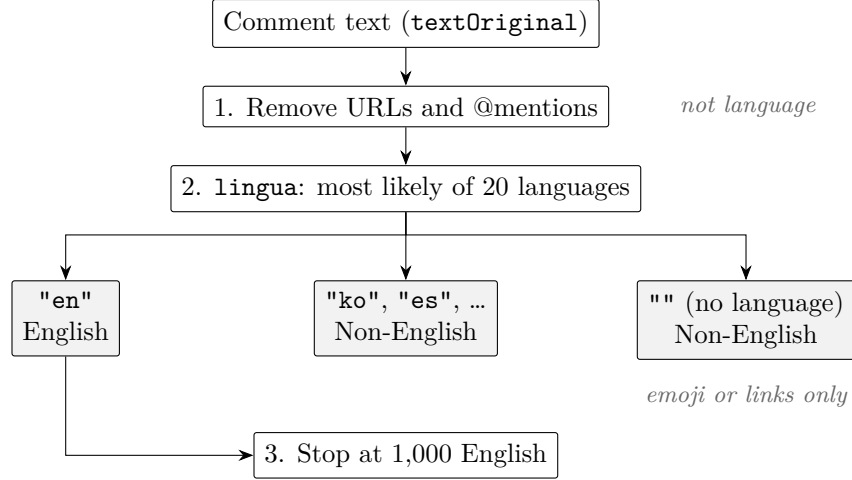


Figure 1: Language label of one comment. Only English comments count toward the stop.

Video	Order	Non-English			Total
		English	(language found)	No language	
BTS <i>Dynamite</i>	relevance	660	413	176	1,249
BTS <i>Dynamite</i>	time	340	289	87	716
BTS <i>Dynamite</i>	total	1,000	702	263	1,965
BLACKPINK <i>Kill This Love</i>	relevance	579	424	105	1,108
BLACKPINK <i>Kill This Love</i>	time	421	434	164	1,019
BLACKPINK <i>Kill This Love</i>	total	1,000	858	269	2,127

Table 3: Fetched comments by order and language. “Time”: newest first.

2 Text cleaning and language split

- Detector: `lingua`, 20 languages common in K-pop comments, picked by hand (`common.LANGUAGES`).
- Why not all 75: short English comments came out as Sotho or Tswana.
- Flow: Figure 1.

Two versions per video:

- **English:** the first 1,000 English comments.
- **Mixed:** all fetched comments, every language.

#	Step	What it does	Why
1	Language	Detect language first.	Gives the English version.
2	Lowercase	All letters lowercase.	As <code>tm</code> .
3	Drop symbols	Drop punctuation, numbers, emoji. Keep every script.	As <code>tm</code> . No cloud font draws emoji.
4	Stopwords	Drop <code>tm</code> English stopwords.	As <code>tm</code> . English only.
5	Short words	Drop words under 3 letters.	<code>tm</code> default <code>wordLengths = c(3, Inf)</code> .
6	No stemming	Keep full words.	The R script loads <code>SnowballC</code> but never calls it.
7	Cloud	Top 150 words, fixed seed.	Same cloud on every run.

Table 4: Cleaning steps from comment to terms.

- No stemming: many stems are not dictionary words (`army` → `armi`, `happy` → `happi`).
- 3-letter minimum: with 2, Spanish and Portuguese words (`de`, `la`, `en`) enter the clouds.

3 BTS *Dynamite* word clouds

- Figures 2 and 3: English, mixed.
- Each cloud: top 150 terms. Word size = term count.



Figure 2: BTS *Dynamite*, English ($n = 1,000$).



Figure 3: BTS *Dynamite*, mixed ($n = 1,965$).

4 BLACKPINK *Kill This Love* word clouds

- Figures 4 and 5: English, mixed.
- Same method as BTS *Dynamite*.



Figure 4: BLACKPINK *Kill This Love*, English ($n = 1,000$).

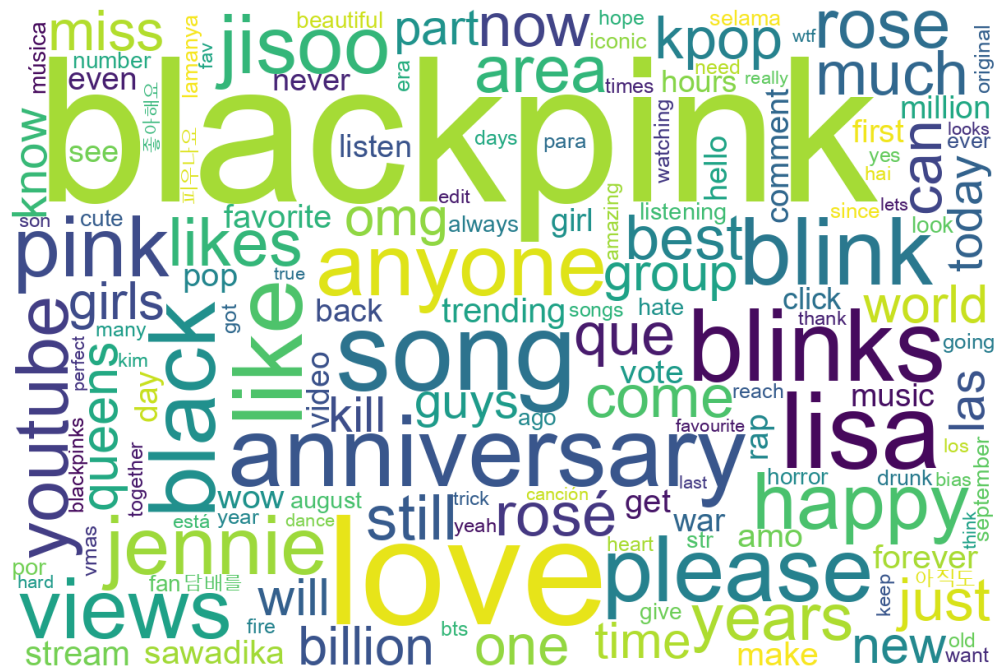


Figure 5: BLACKPINK *Kill This Love*, mixed ($n = 2,127$).

5 Comparison

- All four clouds: Figure 6.
- Engagement: Table 5.
- Top 10 terms: Tables 6 and 7.
- Shared and distinct top 30 terms: Table 8.
- Results only. Discussion: Section 6.

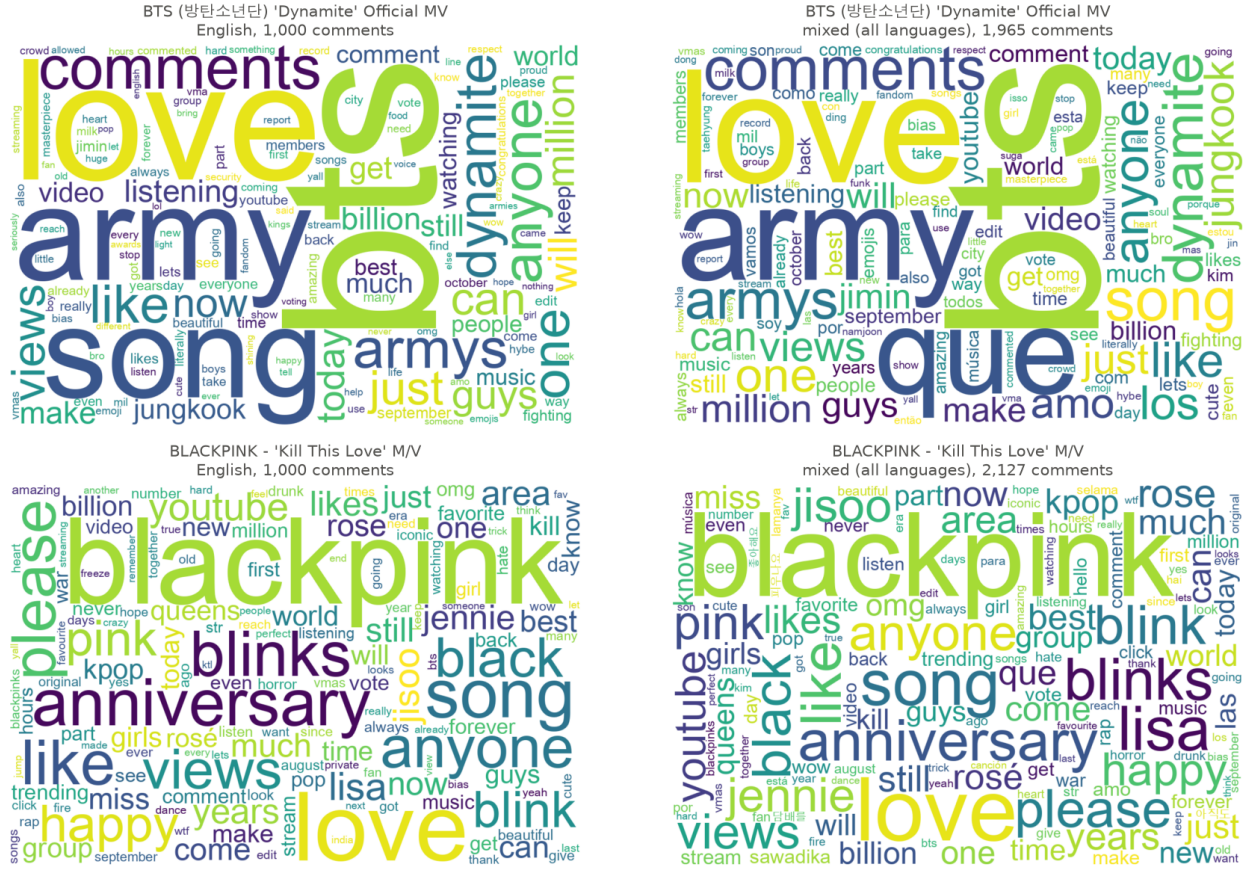


Figure 6: All four clouds. Rows: BTS *Dynamite*, BLACKPINK *Kill This Love*. Columns: English, mixed.

Measure	BTS <i>Dynamite</i>	BLACKPINK <i>Kill This Love</i>
Likes per 1k views	18.47	11.92
Comments per 1k views	7.41	1.11
Share fetched (of YouTube total)	0.0124%	0.0851%
Share English (of YouTube total)	0.0063%	0.0400%

Table 5: Engagement of the two videos.

Rank	BTS	Count	BLACKPINK	Count
1	bts	262	blackpink	351
2	love	148	love	207
3	army	88	song	92
4	song	82	anniversary	83
5	comments	74	blinks	77
6	armys	64	like	76
7	anyone	63	please	72
8	dynamite	61	anyone	66
9	one	58	views	61
10	like	56	black	59

Table 6: Top 10 terms, English.

Rank	BTS	Count	BLACKPINK	Count
1	bts	345	blackpink	448
2	love	161	love	229
3	army	113	song	104
4	que	94	lisa	96
5	comments	93	blinks	85
6	song	92	anniversary	84
7	dynamite	81	like	80
8	armys	80	blink	75
9	anyone	63	black	73
10	like	60	please	72

Table 7: Top 10 terms, mixed (all languages).

Version	In both	Only in BTS <i>Dynamite</i>	Only in BLACKPINK <i>Kill This Love</i>
English	love, song, anyone, one, like, just, views, can, now, still, much	bts, army, comments, armys, dynamite, guys, will, million, today, listening, make, billion, get, jungkook, world, video, comment, watching, people	blackpink, anniversary, blinks, please, black, happy, blink, pink, youtube, lisa, jisoo, years, likes, jennie, area, come, rose, best, miss
mixed (all languages)	love, que, song, anyone, like, views, can, now, youtube, still	bts, army, comments, dynamite, armys, one, los, amo, just, jungkook, million, guys, will, jimin, make, today, video, listening, billion, get	blackpink, lisa, blinks, anniversary, blink, black, please, pink, jisoo, jennie, happy, rose, likes, years, area, best, rosé, much, come, kpop

Table 8: Top 30 terms of each version, split by video.

6 Discussion

Both are Korean K-pop groups: BTS *Dynamite* by a boy group, BLACKPINK *Kill This Love* by a girl group.

Before the analysis: what I expected

- Both have large global fandoms, so many comments in English and other languages, not only Korean.
- Frequent words: group name, song, member names, “love”, fandom name.
- One better-known member gets more mentions.

After: common to both

- Group name, “love”, fandom name and song are in the top 10 (Tables 6 and 7).
- As expected, member names rank high, mostly in the top 30 (Lisa 4th in BLACKPINK mixed).
- Pride in views and fame: **views**, **billion**.
- Fandom names: **army** (BTS), **blinks** (BLACKPINK).
- The mixed comments hold many languages: a global audience.
- **anyone**: the videos are 6 (BTS) and 7 (BLACKPINK) years old. Returning fans ask others who is still watching (“September 2026 anyone?”). Many old fans keep coming back.

After: boy group vs. girl group

- Words show boy vs. girl group: **boys**, **kings** only in the BTS cloud, **girls**, **queens** only in the BLACKPINK cloud.
- BTS fans comment far more. Views are close, but comments per 1,000 views are about 7 times higher (Table 5).
- BTS: **comments**. Fans seem proud of the video’s comment count.
- BTS: **fighting** (the Korean cheer) appears.
- BTS: all members finished military service and tour together again. **comeback** appears, but rarely.
- BTS: recent comments (2026-10-03 and 10-04) reflect the tour’s concert in Bogotá, Colombia.
- BTS: one better-known member holds true. Only Jungkook and Jimin are in the English cloud (2 of 7).
- BLACKPINK: all 4 members are in both clouds. Lisa leads only in mixed.
- BLACKPINK: **anniversary** stands out: the group’s 10th anniversary, mostly in the newest comments.
- BLACKPINK: fans miss the group (**miss**). The members left one agency for their own and work solo.
- BLACKPINK: the fan chant “BLACKPINK in your area” (**area**).

After: English vs. mixed

- **que** (“that”) and **amo** (“I love”), Spanish and Portuguese, appear in both mixed clouds (Table 8): both have Spanish- and Portuguese-speaking fans, more for BTS.
- BLACKPINK: Thai comments appear, Lisa’s home country. BTS has none.

Conclusion

- Both have global fandoms, and old fans keep coming back.
- BTS: the tour shows in recent comments (Bogotá concert). Spanish comments are many, but not more recently.
- BLACKPINK: recent comments center on members and its 10th anniversary.